

# Pranav Kumar Soma

SOFTWARE ENGINEER

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San Francisco Bay Area, CA • Available January 2027

## EDUCATION

### M.S. Computer Science, University of California San Diego

Jan 2026 – Dec 2026

GPA 4.00/4.00 • Jacobs School of Engineering • completed in 1 year

La Jolla, CA

### B.S. Computer Science, University of California San Diego

Sep 2023 – Dec 2025

GPA 3.96/4.00 • Cum Laude, CSE Department Honors, Provost Honors every quarter • completed in 2 years

La Jolla, CA

**Coursework:** Graduate Algorithms, Operating Systems, Distributed Systems, Parallel Computing, Computer Security, Theory of Computation, Software Engineering, Principles of Programming Languages, Data Systems for ML, Computer Vision, Statistical NLP.

## EXPERIENCE

### Software Engineer Intern, ServiceNow

Jun 2026 – Sep 2026

AI Voice Agent Analytics

San Diego, CA

- Shipped a full-stack conversation-history troubleshooting platform unifying **20+ data tables** into overview and per-call drill-down reports; released to production across **24+ enterprise instances** serving **10K+ voice calls/day**
- Cut report load **7s to under 1s** over millions of records (up to 100 interaction records per call) via lazy loading, caching, and an iterative scalability redesign; React, JavaScript, Playwright suites in Jenkins CI/CD
- Owned scope through delivery: authored PRDs and design docs, ran design reviews and sprint demos, presented to a **VP and Senior Director**, and closed high-priority defects and customer escalations solo

### Software Engineer Intern, ServiceNow

Jun 2025 – Sep 2025

AI Voice Agents, IT Service Management • **Technical Skills Award**, top intern of 800+

San Diego, CA

- Owned the **AWS Lex V1 to V2** migration end to end (8 intents, 48+ slots) ahead of end-of-life: **45% lower agent-response latency**, shipped to production and validated live with an enterprise customer. **Return offer**
- Built “Wrap Up,” an AI call-summarization flow (Seismic, custom data brokers, JavaScript client scripts) cutting average call-closing time **6 minutes** across **10K calls/day** per customer telemetry

### AI Research Fellow & Agentic Systems Lead, Laboratory for Emerging Intelligence, UCSD

Nov 2024 – Present

- Lead **6 engineers** on a **Teradata-funded** agentic workbench (**100+** pilot users): **60+ MCP servers** over all **600 UCSD APIs**, a planner/executor split that runs workflows deterministically and degrades gracefully; FastAPI, MongoDB, GCP Cloud Run
- Built and deployed the Python **AI Tutor** to UCSD, SDSU, and Cal Poly Pomona across 10 courses; a controlled study over **600+ students** measured **64% fewer TA overtime hours**

### Engineering Manager, Triton Software Engineering

Sep 2024 – Jun 2025

Software Developer, Sep 2023 – Sep 2024

San Diego, CA

- Led **11 developers and designers** through two delivery cycles of Spay.LA, a MERN fundraising and legislative platform that **raised \$75K**; scoped client requirements into features and owned delivery
- Built the Union Station Homeless Services housing portal in **TypeScript MERN** for LA County’s sole provider of those services: media upload with client-side downsizing, validated intake, state-machine resubmission flows, bulk MongoDB exports

### Computer Vision & Embedded Systems Engineer, Triton Robotics

Sep 2023 – Jun 2024

50-person competitive robotics team

San Diego, CA

- Built real-time motor-control pipelines in **C/C++ over ROS** and Python autonomous-movement CV; implemented conditional **pure-pursuit** planning, PID controllers, and contour-detection tracking

### Honors & M.S. Thesis Researcher, UCSD CSE, Computational Redistricting

Sep 2024 – Present

- Built 3 multi-threaded districting generators over Census 2020 data at **158K to 5.5M blocks/state** (XGBoost + graph partitioning, Sequential Monte Carlo, greedy DP with Kernighan–Lin); Python, pymetis, GeoPandas, Parquet

## SELECTED PROJECTS

- ORCA, AI Music Production** (Best Project, SDTC 2025: 1 of 30 from 200+) • OpenCV score reading, neural source separation, CNN audio-to-score; **98% arrangement accuracy**. Flask, Docker, React
- TSE Constellation** • co-built the org’s published React/TypeScript design system (@tritonse/tse-constellation; Vite, Storybook, atomic design) used org-wide by **100+** engineers

## LEADERSHIP & HONORS

**Co-Founder & President**, Real World AI Network (2026–Present) • **7,000+** student federation, **50+** faculty, 6 schools.

**President**, CSE Society @ UCSD (2025–2026) • **300+** members, **21+** projects, sponsors **2 to 16**, **\$50K+** raised, first **Linux Foundation** hackathon.

**Honors:** ServiceNow Technical Skills Award • Best Project SDTC 2025 • Cum Laude & CSE Department Honors • YC Startup School 2026 • National Merit Finalist • USACO Silver.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, Java, TypeScript, JavaScript, SQL, Bash, ARM Assembly, Swift, Kotlin, HTML/CSS

**Web & Backend:** React, Node.js, Express, FastAPI, Flask, Django REST, MERN, REST APIs, Firebase, Google OAuth; MongoDB, PostgreSQL, MySQL; Playwright, JUnit, Postman, Agile/Scrum, system design, design docs and PRDs

**Systems & Cloud:** Docker, Kubernetes, Jenkins CI/CD, GCP (Cloud Run), AWS (Lex, Lambda), Git, Linux, ROS; distributed, parallel, and real-time systems, finite state machines; PyTorch, XGBoost, OpenCV, LLM fine-tuning, RAG, LLM agents and MCP